The figures

Transcriptomic analyses in neurodevelopment · continuous assessment

This handout exists so that a failed Colab runtime cannot cost you marks. The figures here are the reference output of the notebook: if your own run produced them, answer from your own; if it did not, answer from these. Either way you are marked on what you read out of the figure, not on whether your code ran.

Every figure is numbered exactly as the worksheet refers to it, and each page names the question it belongs to. Figure 5b is the extra panel that Block C produces alongside Figure 5; the worksheet does not ask about it directly, but Q5 is much easier to answer with it in front of you. Figures 1-5 come from simulated data; Figures 6 and 6b are real Visium data.

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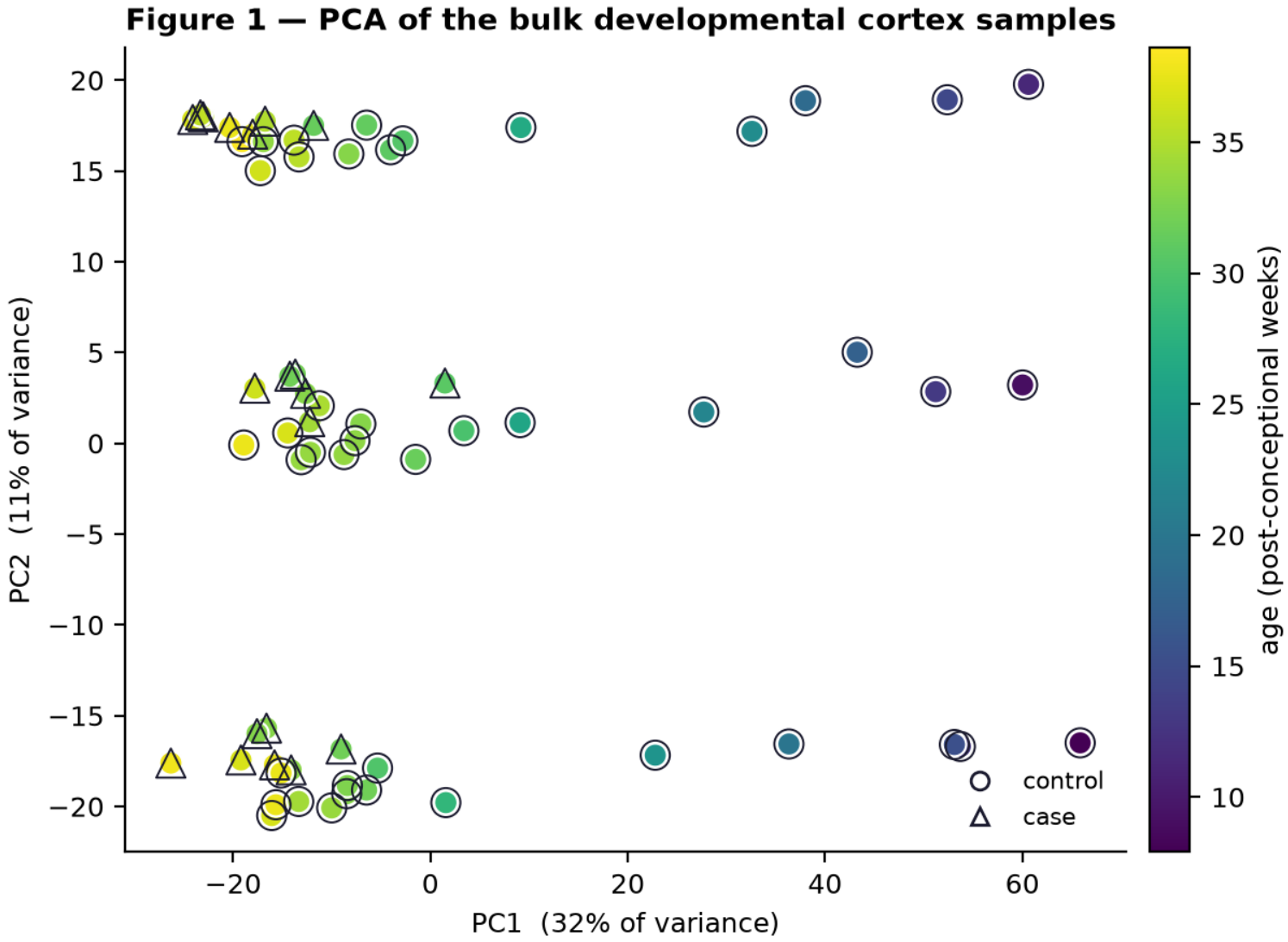
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Figure 1 PCA of the bulk developmental cortex samples

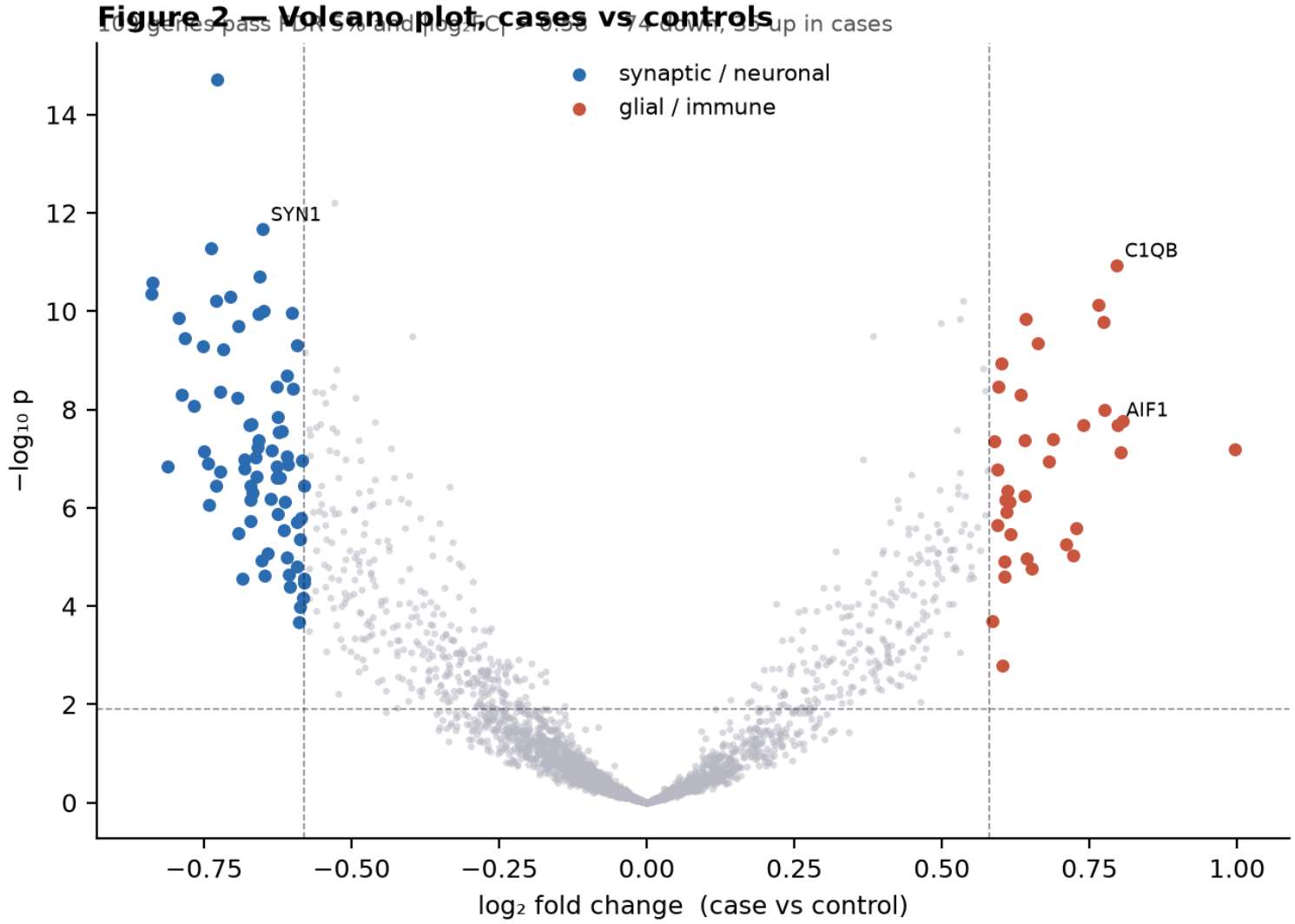
Q1



64 samples. Colour = age in post-conceptional weeks; open circles = controls, triangles = cases. Axis labels carry the share of variance each component explains.

Figure 2 Volcano plot, cases vs controls

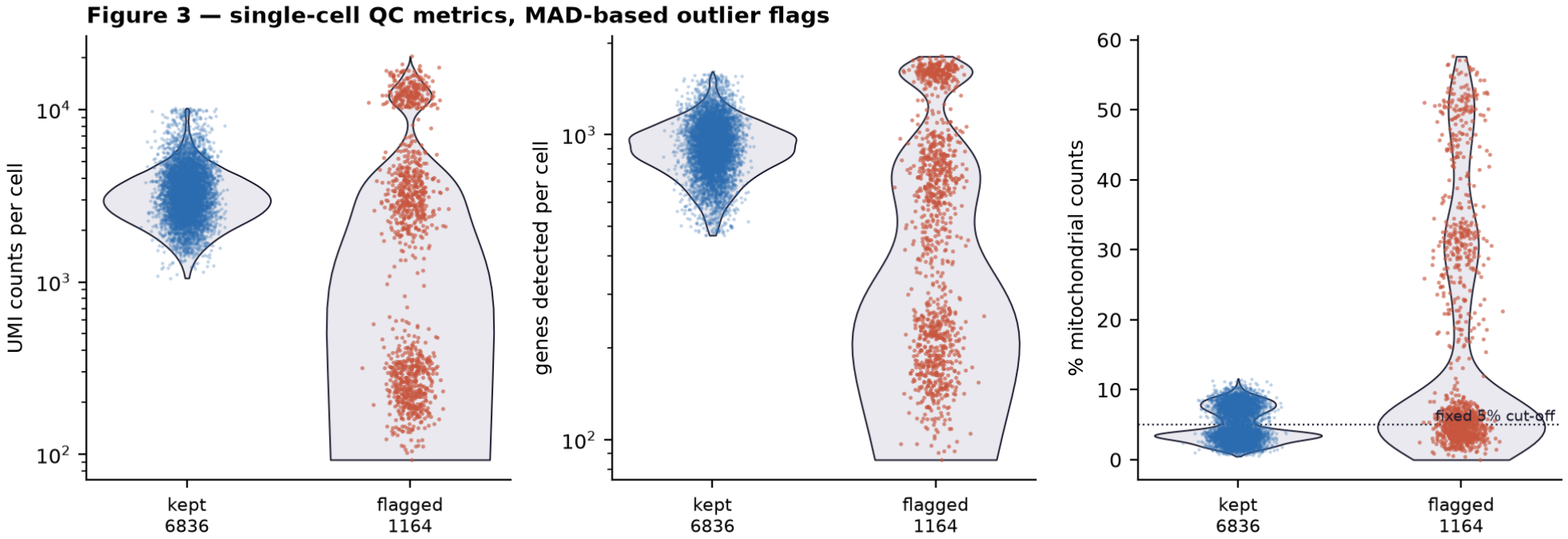
Q2



The 40 age-matched late-fetal samples only. x = log2 fold change, y = -log10 p. Dashed lines mark |log2FC| = 0.58 and the FDR 5% threshold. Coloured points are significant members of the two co-expression modules.

Figure 3 Single-cell QC metrics with MAD-based outlier flags

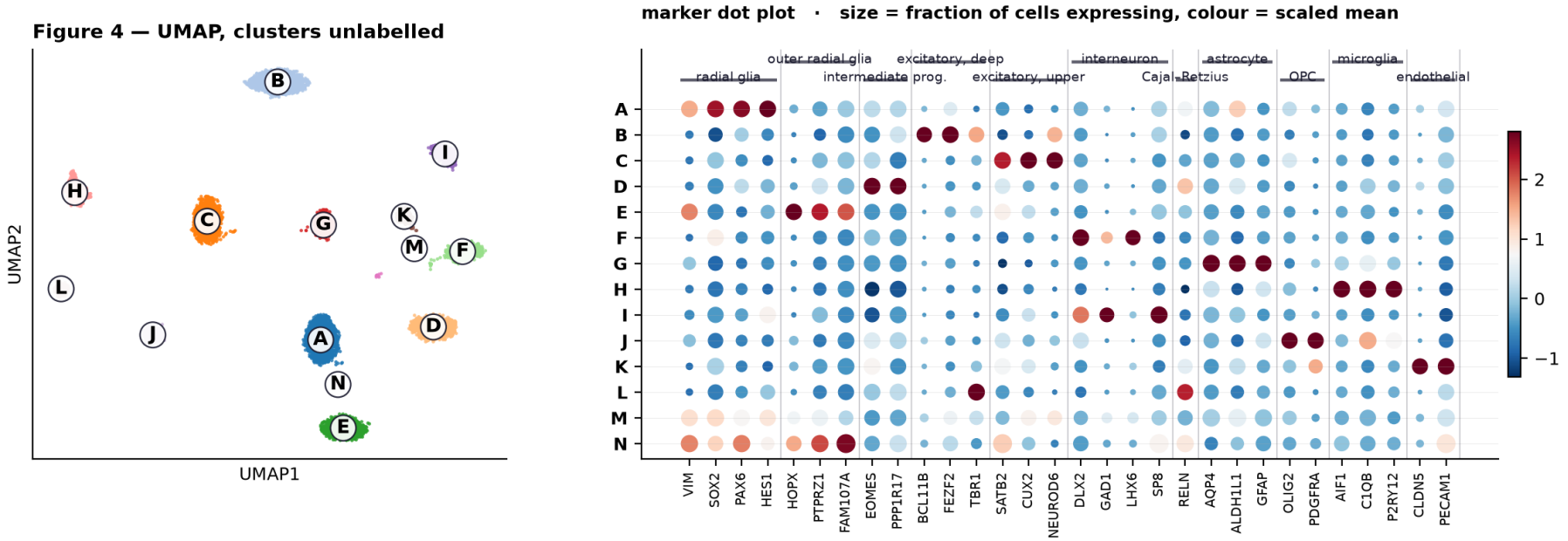
Q3



8,000 raw barcodes, before any filtering. Blue = kept, red = flagged by a median-absolute-deviation rule on log counts, log genes, and mitochondrial fraction. The dotted line on the right panel is a fixed 5% cut-off, for comparison.

Figure 4 UMAP and marker dot plot; clusters are unlabelled

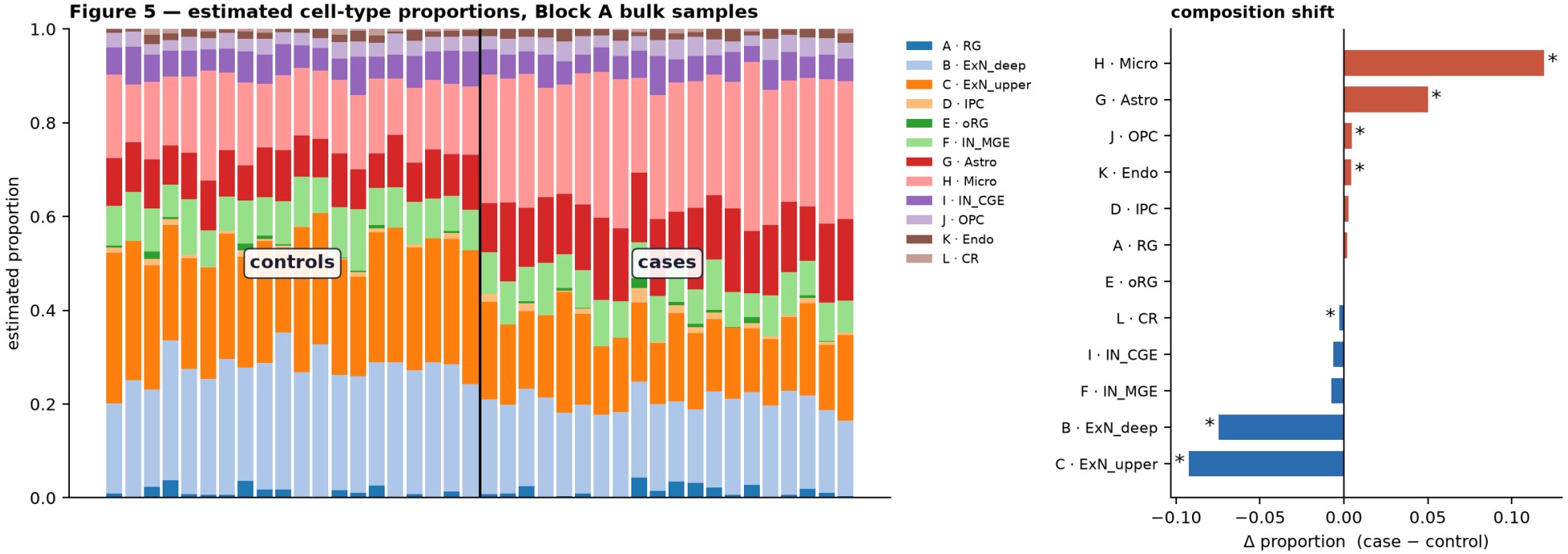
Q4



Leiden clusters, lettered by size. In the dot plot, dot size = the fraction of cells in that cluster expressing the gene, colour = the mean expression scaled across clusters. Marker groups are bracketed above the columns.

Figure 5 Estimated cell-type proportions in the Block A bulk samples

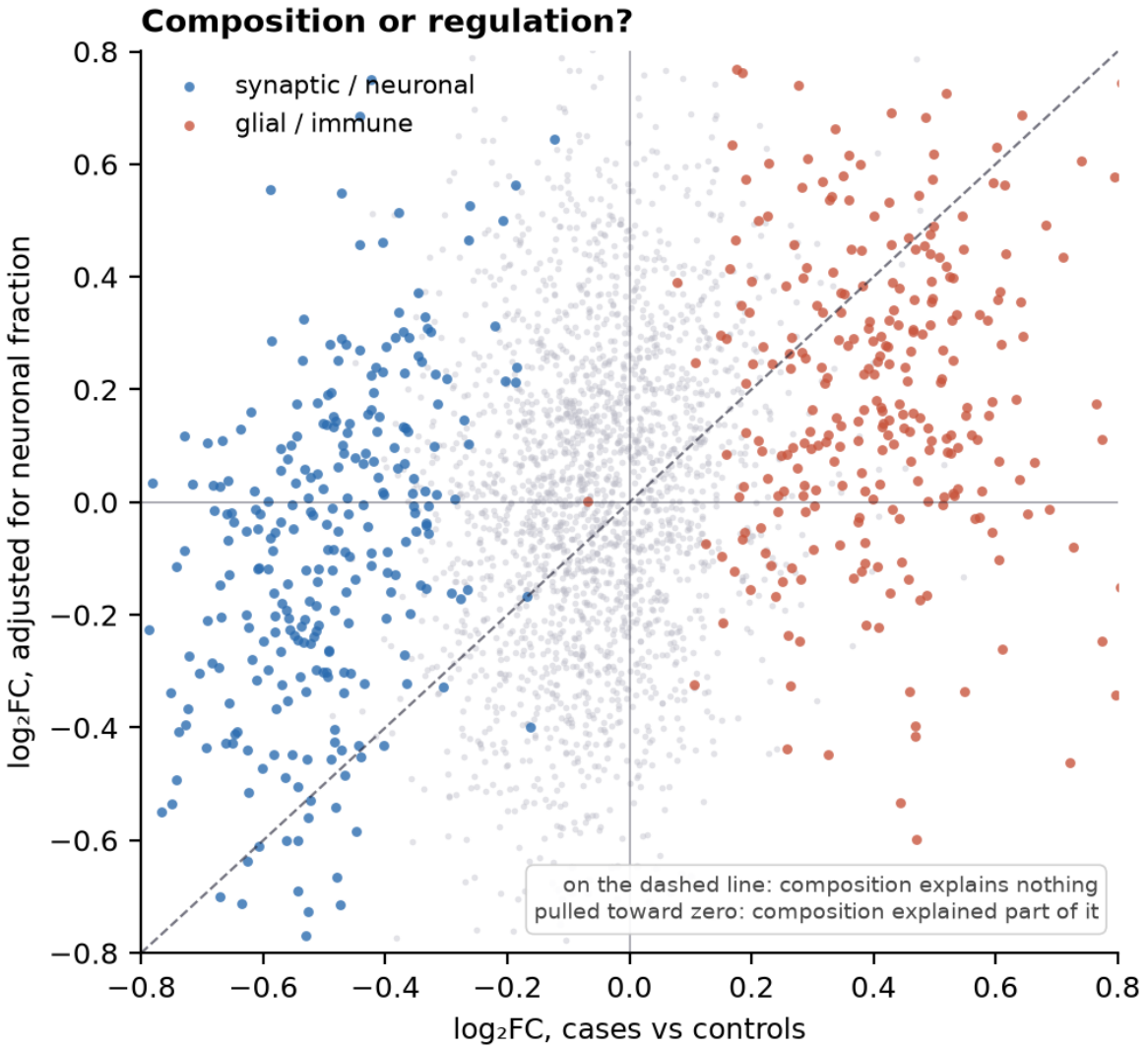
Q5



Each bulk sample of the contrast cohort deconvolved against the Block B cluster signature. Left: the full composition, controls then cases. Right: the difference in mean proportion; * marks $p < 0.05$, Mann-Whitney.

Figure 5b Effect size before and after adjusting for composition

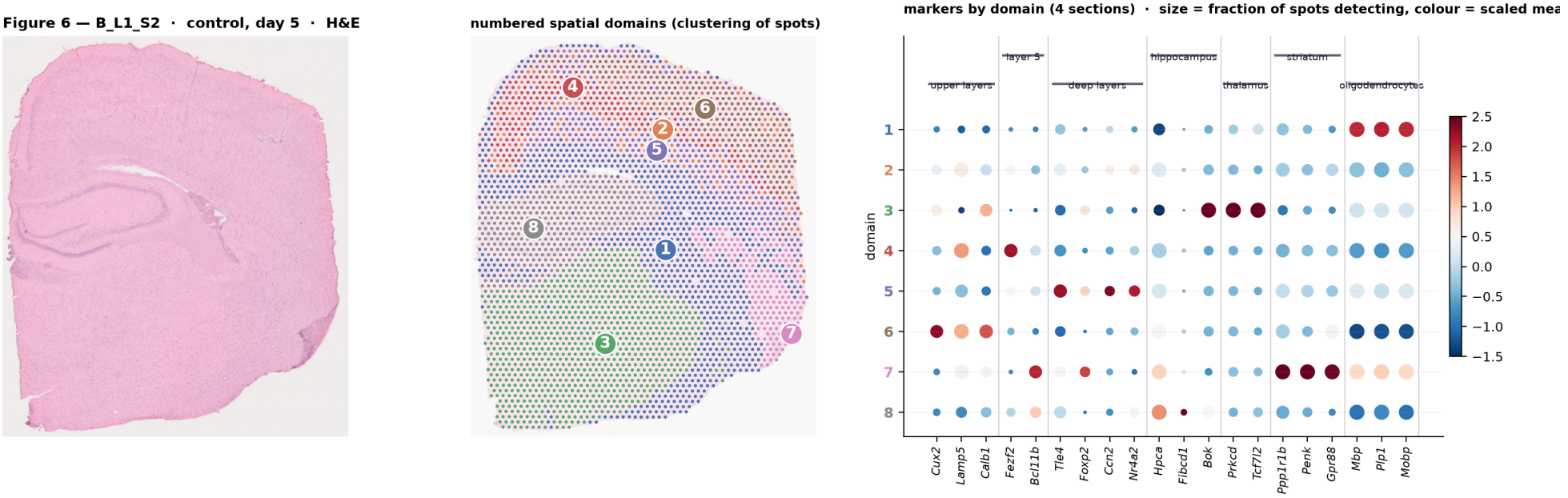
Q5



The same case/control contrast, re-fitted with the estimated neuronal fraction as a covariate. A gene on the dashed diagonal is one that composition does not explain; a gene pulled toward zero is one where it does.

Figure 6 Visium section of rat brain, numbered spatial domains

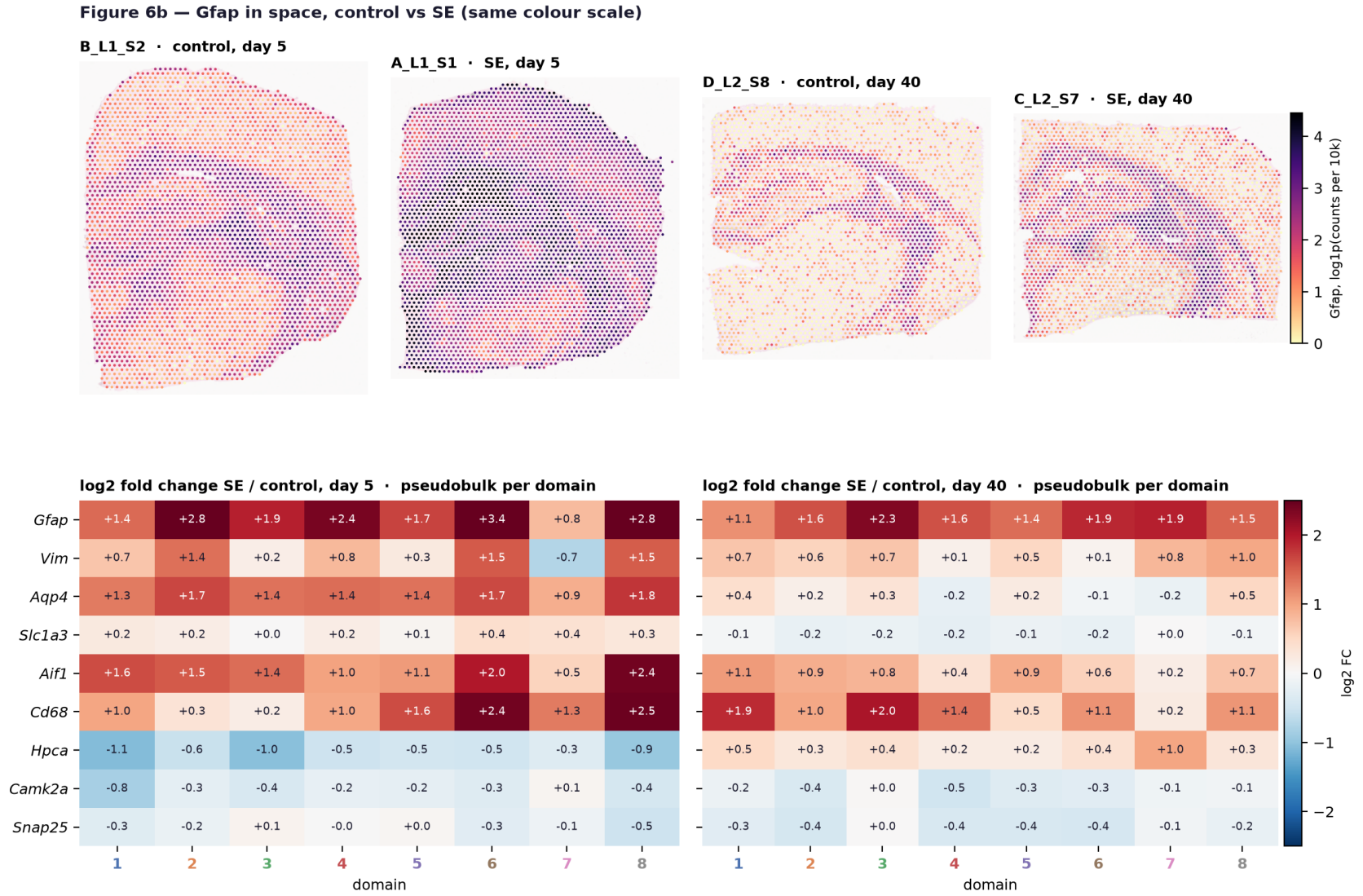
Q6



Real data: a coronal section of a rat hemisphere (control, day 5), about 3,400 spots of 55 um, 100 um apart. Left: H&E.; Middle: the eight spatial domains found by clustering spots, numbered in no particular order. Right: marker genes by domain, over the four sections of the dataset.

Figure 6b Gfap in space, control vs status epilepticus

Q6



Top: Gfap on the four sections (control and SE, 5 and 40 days after SE), one shared colour scale. Bottom: log2 fold change SE / control per domain, every spot of a domain summed into one pseudobulk profile, for reactive, astrocytic, microglial and neuronal genes.